

CS485 — Data Science and Applications

Spring 2026

Homework 2

Deep Learning Foundations · MLP Training · Model Analysis

Submission Deadline: May 28, 2026

Objective

In this homework you will extend the analysis from Homework 1 by replacing the linear regression baseline with a **Multi-Layer Perceptron (MLP)**. You will apply the concepts covered in Lectures 11–15 to understand how neural networks are trained, how hyper-parameter choices affect generalisation, and how the predictions of a deep model compare to the simpler models you already built.

Dataset continuity is required. You must reuse the *same* Multimodal Universe dataset, modalities, and target variable as in Homework 1. No new data source may be introduced. The train/validation/test split defined in Part 0 must be carried forward consistently across all parts.

Data

Continue working with the **Multimodal Universe** dataset you selected in Homework 1:

<https://github.com/MultimodalUniverse/MultimodalUniverse>

Support code and notebooks are available at:

<https://github.com/gtsagkatakis/Data-Science-and-Applications-2026>

Your data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and target vector $\mathbf{y} \in \mathbb{R}^n$ remain the same. The pre-processing and feature choices from Homework 1 carry over; changes can be made but must be documented and justified.

Part 0 — Data Preparation & HW1 Bridge

Goal: Establish a reproducible pipeline and connect findings from Homework 1 to the modelling decisions made here.

3.1 Split & Normalise

- Divide your dataset into **training**, **validation**, and **test** sets using a 70 / 15 / 15 split. Fix a random seed and document it at the top of your notebook.
- Fit a `StandardScaler` on the *training set only* and apply the same transformation to the validation and test sets.
- **Discussion:** Why is it incorrect to fit the scaler on the full dataset before splitting? What information would leak, and how could it affect reported performance?

3.2 HW1 Insight

- In a Markdown cell, state **one concrete finding** from your Homework 1 EDA or regression analysis, for example, a feature with a heavily skewed distribution, a modality that dominated prediction, or non-Gaussian residuals.
- Explain in 2–3 sentences if/how this finding motivates a specific choice you make in Homework 2 (architecture, pre-processing, or analysis strategy).

Part 1 — Baseline MLP: Architecture & Training

Goal: Build and train an MLP on your regression task and establish a direct comparison with the linear model from Homework 1.

4.1 Architecture Definition

- Define an MLP with **at least two hidden layers** using PyTorch or sklearn.
- Document design choice in a Markdown cell: number of layers, neurons per layer, activation function(s), output layer design, and total trainable parameter count.
- **Discussion:** Justify your choices. Why did you select this depth and width? What activation function did you use, and why is it appropriate for a regression task?

4.2 Training Loop

- Train using the Mean Squared Error loss:

$$\mathcal{L}(w) = \frac{1}{n} \sum_{i=1}^n (f(x_i; w) - y_i)^2$$

- Record the loss at every epoch and **plot training loss and validation loss on the same axes**.
- **Discussion:** At which epoch does the model appear to converge? Do the two curves diverge at any point? What does divergence indicate?

4.3 Evaluation & Comparison

- Report **MSE** and R^2 on the held-out test set.
- Present a **comparison table** with at least the following rows:

Model	Test MSE	Test R^2
Linear regression (HW1)		
Baseline MLP (this work)		

- Produce a **predicted-vs-actual scatter plot** for the MLP. Compare it visually to the equivalent plot from Homework 1.

Part 2 — Training Dynamics

Goal: Understand how hyper-parameter choices affect the training process and generalisation, building on the learning-rate experiments from Homework 1.

5.1 Learning Rate Sensitivity

- Train the same architecture with **at least three different learning rates** α . Plot all training loss curves on the same axes.
- **Discussion:** How does α affect convergence speed and stability? Compare your observations to the gradient descent experiments in Homework 1 — what is similar, and what is different when training a multi-layer network?

5.2 Optimiser Comparison

- Train with **SGD** and **Adam** using the same architecture and number of epochs. Report the final validation loss and test MSE for each.
- **Discussion:** Which optimiser converged faster? Which achieved lower final loss? What structural difference between SGD and Adam explains this behaviour?

5.3 Early Stopping

- Implement **early stopping** manually: halt training when the validation loss has not improved for N consecutive epochs (choose and justify N).
- Report the epoch at which training stopped and compare the resulting test MSE to the fully-trained model from Section 1.2.
- **Discussion:** Did early stopping improve or worsen test performance? What does this tell you about overfitting in your model?

Part 3 — Feature & Error Analysis

Goal: Connect model performance back to data structure, exploiting the insights established in Homework 1.

6.1 Residual Analysis

- Compute residuals from your best MLP:

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$$

- Overlay the KDE of the MLP residuals and the KDE of the Homework 1 linear model residuals on the **same plot**.
- **Discussion:** What changed between the two models? Does the MLP residual distribution more closely resemble a Gaussian? Are there systematic biases that persist?

6.2 Error Inspection

- Identify the **five samples with the largest absolute residuals** from your best MLP. Print their feature values and their true and predicted target values.
- Cross-reference with the outliers identified in your Homework 1 EDA (histograms, scatter plots).
- **Discussion:** Are the hardest samples the same outliers flagged in Homework 1? What do they have in common? What does this suggest about the limits of the model?

Deliverables

Submit a single **Jupyter Notebook** (.ipynb) containing:

- All code, clearly commented and organised by Part/Section.
 - All plots and numerical results rendered inline.
 - Written discussion cells for every **Discussion** item above (use Markdown cells, not code comments).
 - A short introduction cell (**Part 0**) documenting your dataset, the random seed used, and the HW1 insight that motivates your HW2 approach.
 - The comparison tables from Sections 1.3 and 3.2 rendered as formatted output (e.g. a **pandas** DataFrame or printed table).
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